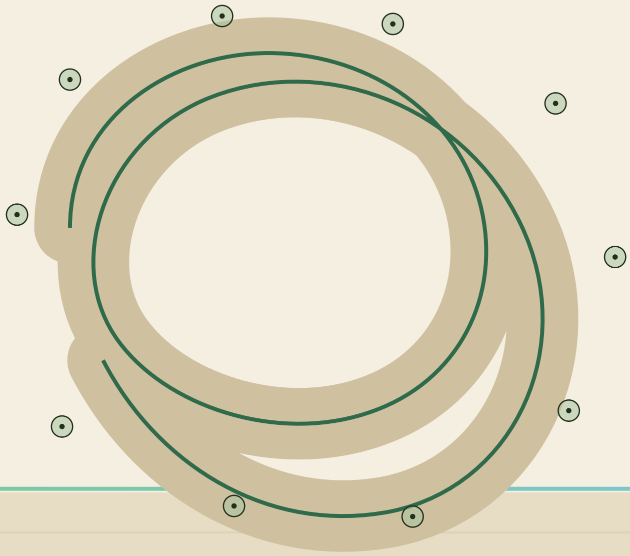


nabta WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY · AI PARK DESIGN CHALLENGE · DRAFT COMPETITION ENTRY

D·01

Executive Summary & Full Presentation Outline

№01 Executive Summary

№14 Full Presentation

«نبتة» NABTA — Wellness Park

`Concept: The Living Shade Loop` `Deliverable 1 · Executive Summary` `Al Safa 2 · 15,001 m² · AED 35M`

VISION

> ***A park where comfort makes healthy habits possible.*** *Nabta ("sprout") reimagines Al Safa 2 as a wellness park organized around an 858 m continuous shaded network — a figure-eight threaded through a perimeter ring, roughly 1 km including the wadi spur — part canopy forest, part AI-form-found pergola, comfortable enough to walk, jog and gather from morning school-runs to midnight strolls, eleven months a year. The network threads ten "garden rooms," so a 15,001 m² neighborhood parcel behaves like a much larger park: every lap moves the body through plaza, play, fitness, picnic groves, a sensory wadi garden and quiet restoration. AI shaped every decision — from where shadows fall in August to where a seven-year-old runs at 5pm — while our design team made every call.*

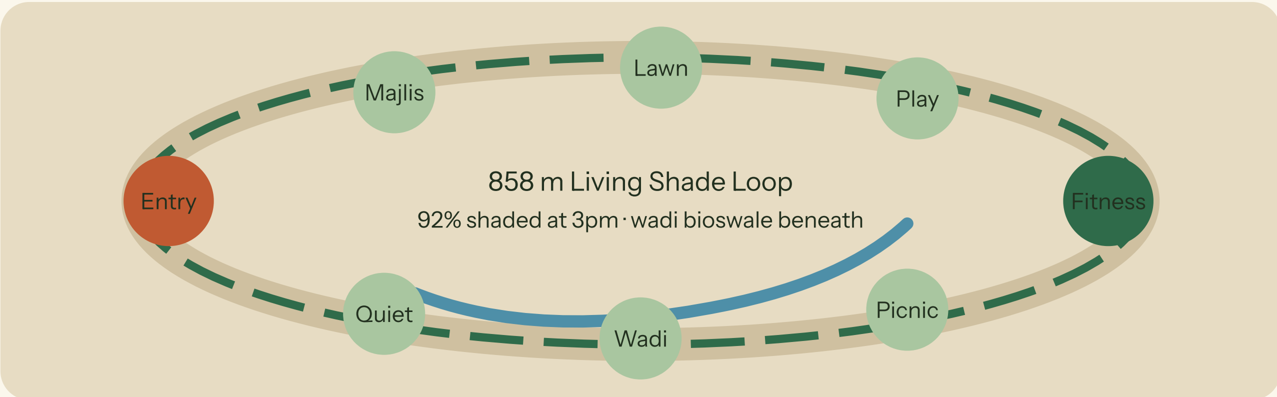
FIVE KEY INNOVATIONS

1. **The Living Shade Loop as health infrastructure.** The 858 m walking-jogging network — a 398 m figure-eight inside a 460 m perimeter ring — is the park's climatic spine: ghaf, sidr and date-palm canopies alternate with parametric pergola segments placed by an open genetic-algorithm search against hourly solar simulation — **92% of the loop shaded at 3pm in summer**, cutting perceived temperature by **4.5°C (UTCI)** so daily movement stays safe and habitual.
2. **Agents before architecture.** An agent-based simulation across six AI-assisted personas (seniors, dawn runners, young children, teens, People of Determination, delivery riders) stress-tested circulation, desire lines and peak crowding on the true parcel geometry — and projected **~1,800 daily visits** — before a single path was drawn.
3. **Generative options, human-judged.** A genetic algorithm searched thousands of shade-element placements (1.7× better than random — published open-source), and four full concept variants were scored against the site's measured sun and wind; the team hybridized the winners, documenting every override.
4. **The Wadi Water Loop.** A bioswale garden harvests AC condensate from adjacent buildings plus TSE supply, feeding soil-moisture-driven irrigation of a 100% native palette of **~450 trees** — **41% less irrigation water** than a conventional lawn-based park of equal size.
5. **A park that measures wellbeing.** Anonymous footfall counters, UTCI comfort sensors, adaptive lighting and irrigation telemetry feed a public dashboard reporting active-minutes on the loop, comfort hours per day and water saved — giving Dubai Municipality auditable health and sustainability KPIs, not just maintenance data.

HEADLINE METRICS

- **92%** of the 858 m loop network shaded at 3pm, summer (simulation verifies 91.6%; random placement scores 18.3%)
- **~1,800** daily visits projected (agent simulation); **ten** garden rooms, play zoned 2–5, 6–12, 13–17

- **−4.5°C** perceived temperature (UTCI) along the loop vs the existing park
- **41%** reduction in irrigation water vs a conventional lawn-based park
- **100%** of primary routes step-free, tactile-guided, ≤1:21 gradient
- **~450** native trees (ghaf, sidr, date palm); year-5 canopy-closure target with pergolas bridging the early years
- **18-hour** activation day; dawn fitness to evening family strolls under adaptive lighting
- **AED 35.0M** fully allocated cost plan — 6% contingency inside the cap



CONCEPT DIAGRAM: THE SHADED LOOP THREADS THE GARDEN ROOMS; THE WADI BIOSWALE (BLUE) CROSSES THE INTERIOR. SELECTED ROOMS LABELED.

DELIVERABLE 14 — FULL PROJECT PRESENTATION (SLIDE OUTLINE)

#	SLIDE TITLE	CONTENTS
1	NABTA — Wellness Park: The Living Shade Loop	Title, hero aerial render at golden hour, team identity, one-line vision.
2	Al Safa 2 Today	Official as-built parcel: irregular 16-vertex boundary, 15,001 m², ~120 × 178 m envelope, street edge to the south, grid rotated ~47° from north; measured satellite vegetation and heat, solar and wind analysis, catchment description; on-site tree audit planned before design freeze.
3	Who the Park Serves	Six AI-assisted personas; wellbeing-needs matrix (movement, restoration, social connection); day-in-the-life timelines; ~1,800 daily visits projected.
4	The Big Idea	Shade-first thesis; loop + ten garden rooms diagram; precedent of the sikka and the wadi.
5	AI Workflow Overview	End-to-end pipeline map: analysis → personas → agent simulation → generative options → environmental scoring → human selection.
6	Four Variants, One Winner	Concept variants A–D scored on the site's real sun and wind (satellite-verified, variant C wins at 0.72); shade-placement optimizer results (31.8% vs 18.3% random); how the team hybridized the strongest ideas.
7	Masterplan	Illustrated 1:500 plan on the as-built parcel; 398 m figure-eight + 460 m perimeter ring path network; program areas and m² breakdown; south street-edge entries, service access, wayfinding logic.
8	Microclimate Proof	UTCI heatmaps before/after; 3pm summer shade animation stills; −4.5°C result and pergola form-finding.
9	Garden Rooms Tour I	Entry plaza, majlis plaza, event lawn, café terrace — renders, capacities, event configurations.

#	SLIDE TITLE	CONTENTS
10	Garden Rooms Tour II	Inclusive play dune (by age group), teen zone, fitness grove, picnic groves, sensory wadi, quiet garden; accessibility features throughout.
11	Water & Planting	~450 native trees (ghaf, sidr, date palm, desert grasses); wadi bioswale and condensate loop; 41% irrigation saving math.
12	Measuring Wellbeing	Smart-ops layer: anonymous footfall, comfort sensing, adaptive lighting, irrigation telemetry; public dashboard of active-minutes and comfort hours.
13	Cost, Phasing, Operations	AED 35.0M fully allocated elemental cost plan (softscape 23%, shade structures 19%, hardscape 13%, wadi & water 10%, buildings 9%, play 7%, sports/lighting/wayfinding 12%, contingency 6%); 18-month, 3-phase build with the park partially open from month 12; O&M model per DM standards.
14	A Lap of NABTA	one-minute concept film (current cut 56 s; final 60 s planned with 3D-reconstruction footage); closing metrics strip; call to action.

COST PLAN — AED 35,000,000 IMPLEMENTATION BUDGET, FULLY ALLOCATED

ELEMENT	AED M	%
Softscape & planting — ~450 trees, understorey, soil, irrigation network	8.2	23%
Shade structures — loop pergolas, gateway, sails	6.8	19%
Hardscape — 858 m continuous loop (398 m figure-eight + 460 m perimeter ring), paths, plazas, seat walls	4.6	13%
Wadi bioswale & water systems — TSE + condensate network, pumps, treatment	3.6	10%
Buildings — café kiosk, restrooms (incl. Changing Places), service yard	3.2	9%
Play equipment — inclusive dune, carousel, nets	2.6	7%
Sports & fitness — court, calisthenics, surfaces	1.6	5%
Lighting & smart systems — adaptive dark-sky, sensors, irrigation controls	1.8	5%
Wayfinding, signage & FF&E	0.6	2%
Preliminaries & contingency	2.0	6%
Total *(percentages rounded)*	**35.0**	**100%**

Cost risk concentrates in two lines — pergola steel and the TSE connection — and the contingency is sized against them. Every other element uses standard regional rates for neighborhood-park construction.

PHASING & OPERATIONS — 18 MONTHS, THREE PHASES, PARK NEVER FULLY CLOSED

- **Phase 1 · months 1–6 — Ground & water.** Site preparation, earthworks, the wadi bioswale, TSE + condensate connections and the irrigation mains. The eastern strip remains open to the neighborhood throughout.
- **Phase 2 · months 7–12 — Loop & structures.** The 858 m path network, pergolas and gateway, buildings and courts. **The park partially opens at month 12** — loop, entry plaza and play dune in use while planting continues.
- **Phase 3 · months 13–18 — Planting & fit-out.** ~450 trees and understorey planted into the cooler season for establishment, lighting and smart-ops commissioning, FF&E, snagging and handover.
- **Maintenance target:** AED 28/m²/yr ≈ AED 420k/yr (vs AED 50–60/m² for a conventional turf park) — achieved by native planting, drip irrigation and low-mow groundcovers.

- **Concession income:** café + event-lawn cart pads estimated at AED 250–350k/yr — offsetting roughly 70% of maintenance.
- **Water cost:** zero potable — 53 m³/day TSE + 9 m³/day recovered condensate at peak demand.

All AI outputs in this submission were generated, curated and verified under design-team direction; final design decisions are human-made and fully documented in the AI-usage log (Deliverable Package 4).